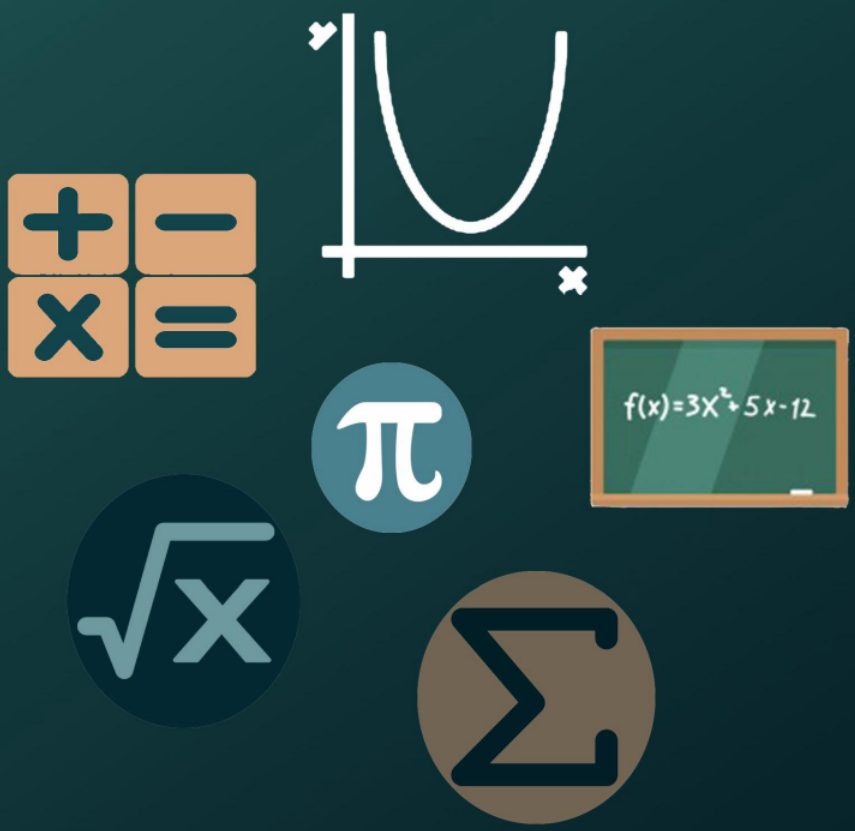
MATHEMATICS

Limit



SHEET

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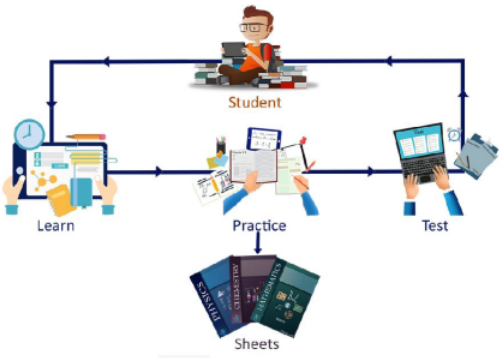
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topics	EX - 1	EX - 1A	EX - 2	EX - 2A
1	Concept of Limit				
2	Existence of Limit	1 - 4	1 - 2, 7 - 9, 25	1 - 3	
3	0/0 Form - Factorisation	5 - 7	3 - 4, 16 - 17, 26		19 - 20
4	0/0 Form - Rationalization	8 - 13	22, 24		1
5	0/0 Form - Standardization	14			
6	Infinity by Infinity Form	15 - 20	5, 13, 18 - 19, 23, 27	4 - 5	21
7	Zero into Infinity and Infinity upon Infinity Form				
8	Limit of Trigonometric Functions - Part 1	21 - 25	14, 28	6 - 10	12, 16 -18, 26 - 28
9	Limit of Trigonometric Functions - Part 2	26 - 29		11 - 16	2, 13
10	Limit Using Series Expansion	30 - 33	20 - 21	17 - 21	3, 22
11	Limit of Exponential Functions	34	29-30	22 - 23	4
12	Limit of Logarithmic Functions	35	31-32		29
13	(1)^Infinity Form	36-39	6, 15, 33	24 - 25	30
14	Built in Limit				
15	Geometrical Limit				23 - 24 , 31 - 32
16	Limit of Composite Functions				14, 25
17	L'Hospital Rule	40-47	34	26 - 33	5
18	0^0 & Infinity^0 Form				
19	Sandwich Theorem			34 - 36	9 - 11, 33
20	After complete chapter		10 - 12		15, 6-8, 34

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Existence of Limit

Q1. If $f(x)=\begin{cases}4x, & x<0\\ 1, & x=0\\ 3x^2, & x>0\end{cases}$ then $\lim\limits_{x\to 0} f(x)$ equals -

(A) 0 (B) 1

(C) 3 (D) Does not exist

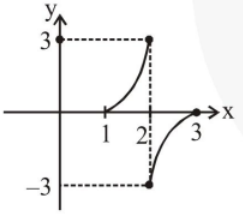
Q2. If $f(x)=\begin{cases}\dfrac{[x]^2+\sin[x]}{[x]} & \text{for } [x]\neq 0\\ 0 & \text{for } [x]=0\end{cases}$

where [x] denotes the greatest integer function, then $\lim\limits_{x\to 0} f(x)$, is -

(A) 1 (B) 0

(C) –1 (D) Does not exist

Q3. If graph of f(x) is given as :-



Then $\lim\limits_{x\to 2} f(x)$ is :-

(A) 0 (B) 3

(C) –3 (D) Doesn't exist

Q4. If $f: R \to R$ is defined by $f(x) = [x-3] + |x-4|$ for $x \in R$;

(Where [.] is G.I.F.) then $\lim\limits_{x\to 3^-} f(x) =$

(A) –2 (B) –1

(C) 0 (D) 1

0/0 Form - Factorisation

Q5. The value of $\lim\limits_{x\to 2}\left(\dfrac{2x^2-7x+6}{5x^2-11x+2}\right)$ is-

(A) 1/2 (B) 1/4

(C) 1/9 (D) Does not exist

Q6. The value of $\lim\limits_{x\to 64}\left(\dfrac{\sqrt{x}-8}{\sqrt[3]{x}-4}\right)$ is-

(A) 1 (B) 2

(C) 3 (D) Does not exist

Q7. $\lim\limits_{x\to -1}\left(\dfrac{x^2-1}{x^2+3x+2}\right)$ is equal to-

(A) – 2 (B) 1/2

(C) 0 (D) 1

0/0 Form - Rationalization

Q8. The value of $\lim\limits_{x\to 5}\dfrac{1-\sqrt{x-4}}{x-5}$ is-

(A) –1/2 (B) 3/4

(C) 1 (D) 3/8

Q9. $\lim\limits_{x\to \pi/4}\left(\dfrac{1-\tan x}{1-\sqrt{2}\sin x}\right)$ is equal to -

(A) 0 (B) 1

(C) –2 (D) 2

Q10. $\lim\limits_{x\to 2a}\dfrac{\sqrt{x-2a}+\sqrt{x}-\sqrt{2a}}{\sqrt{x^2-4a^2}}$ is equal to -

(A) $\dfrac{1}{\sqrt{a}}$ (B) $\dfrac{1}{2\sqrt{a}}$

(C) $-\dfrac{1}{\sqrt{a}}$ (D) $-\dfrac{1}{2\sqrt{a}}$

Q11. $\lim_{x\to 2}\left(\frac{x^2-4}{\sqrt{x+2}-\sqrt{3x-2}}\right)$ is equal -

(A) –8 (B) 8 (C) 4 (D) 0

Q12. The value of $\lim_{x\to 4}\frac{\sqrt{1+\sqrt{x+5}}-2}{4-x}$ is-

(A) $-\frac{1}{2}$ (B) $-\frac{1}{4}$

(C) $-\frac{1}{6}$ (D) $-\frac{1}{24}$

Q13. $\lim_{x\to 0}\frac{\sqrt{x^2+1}-1}{\sqrt{x^2+16}-4}=$

(A) 1 (B) 2 (C) 3 (D) 4

0/0 Form - Standardization

Q14. $\lim_{h\to 0}\left(\frac{(x+h)^{1/3}-x^{1/3}}{h}\right)$ equals -

(A) $\frac{1}{3}x^{2/3}$ (B) $\frac{1}{3}x^{-2/3}$

(C) $\frac{1}{3}x^{1/3}$ (D) $3x^{-2/3}$

Infinity by Infinity Form

Q15. $\lim_{n\to\infty}\frac{3^{n+1}+4^{n+1}}{3^n+4^n}$ equals-

(A) 3 (B) 1

(C) 0 (D) 4

Q16. The value of $\lim_{x\to\infty}x^{3/2}(\sqrt{x^3+1}-\sqrt{x^3-1})$ is-

(A) 1 (B) – 1

(C) 0 (D) None of these

Q17. The value of $\lim_{n\to\infty}\frac{\frac{1}{2}+1+\frac{3}{2}+\ldots+\frac{n}{2}}{25n^2+n+3}$ is -

(A) 0 (B) 1/100

(C) ∞ (D) None of these

Q18. $\lim_{x\to\infty}\frac{\sqrt{x^2+1}-\sqrt[3]{x^2+1}}{\sqrt[4]{x^4+1}-\sqrt[5]{x^4-1}}$ is equal to -

(A) 1 (B) –1

(C) 0 (D) none of these

Q19. If $0<a<b$ then $\lim_{n\to\infty}\left(a^n+b^n\right)^{1/n}=$

(A) a (B) b

(C) a/b (D) e

Q20. $\lim_{n\to\infty}\frac{4n+(-1)^n}{5n+(-1)^n}$ is equal to :-

(A) $\frac{4}{5}$ (B) $\frac{5}{4}$

(C) $\frac{3}{4}$ (D) None of these

Limit of Trigonometric Functions P1

Q21. $\lim_{x\to 0}\frac{\sin^2 2x}{x^2}$ equals-

(A) 1 (B) 2

(C) 4 (D) Does not exist

Q22. $\lim_{x\to 0}\frac{(1-\cos 2x)\sin 5x}{x^2\sin 3x}$ equals-

(A) 1/3 (B) 1

(C) 10/3 (D) 5/3

Q23. The value of $\lim_{x\to\pi/2}\frac{1-\sin^3 x}{\cos^2 x}$ is -

(A) $-\frac{3}{2}$ (B) $\frac{3}{2}$

(C) 1 (D) 0

Q24. The value of $\lim_{x\to 0}\frac{\sqrt{2}-\sqrt{1+\cos x}}{\sin^2 x}$ is -

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{8\sqrt{2}}$

(C) $\frac{1}{4\sqrt{2}}$ (D) $-\frac{1}{4\sqrt{2}}$

Q25. $\lim_{x\to 0}\frac{1-\cos^3 x+\sin^3 x+\ell n\left(1+x^3\right)+\ell n(1+\cos x)}{x^2-1+2\cos^2 x+\tan^4 x+\sin^3 x}$ is :-

(A) $\frac{3}{4}$ (B) $\ell n2$

(C) $\frac{\ell n2}{4}$ (D) $\frac{3}{2}$

Limit of Trigonometric Functions P2

Q26. $\lim_{x\to\pi/2}\frac{\left(1-\tan\left(\frac{x}{2}\right)\right)(1-\sin x)}{\left(1+\tan\left(\frac{x}{2}\right)\right)(\pi-2x)^3}$ is -

(A) ∞ (B) $\frac{1}{8}$

(C) 0 (D) $\frac{1}{32}$

Q27. $\lim_{x\to 4}\left(\frac{\sqrt{1-\cos\{2(x-4)\}}}{x-4}\right)$ equals -

(A) L.H.L. $=-\sqrt{2}$ & RH.L $=\sqrt{2}$

(B) L.H.L. $=-\sqrt{4}$ & R.H.L. $=\sqrt{4}$

(C) L.H.L. $=-4$ & R.H.L. $=4$

(D) Limitexist & equalto $\sqrt{2}$

Q28. The value of $\lim_{x\to 0}\left(\left[\frac{100x}{\sin x}\right]+\left[\frac{99\sin x}{x}\right]\right)$ is :-

(where [.] is greatest integer function).

(A) 199 (B) 198

(C) 0 (D) does not exist

Q29. $\lim_{x\to 0}\frac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$?

(A) $\frac{1}{2}$ (B) –2

(C) 2 (D) $-\frac{1}{2}$

Limit Using Series Expansion

Q30. $\lim_{x\to 0}\left(\frac{\sin x\sin^{-1}x-x^2}{x^6}\right)$ equals-

(A) 1/3 (B) 3/8

(C) 1/18 (D) Does not exist

Q31. $\lim_{x\to 0}\frac{e^x+\log[(1-x)/e]}{\tan x-x}$ equals-

(A) 1 (B) –1

(C) 1/2 (D) –1/2

Q32. $\lim_{x\to 0}\frac{(1+x)^{1/x}-e}{x}$ equals

(A) e (B) e/2

(C) – e (D) – e/2

Q33. The integer n for which

$\lim_{x\to 0}\frac{(\cos x-1)\left(\cos x-e^x\right)}{x^n}$

is a finite non zero number is :-

(A) 1 (B) 2

(C) 3 (D) 4

Limit of Exponential Functions

Q34. $\lim_{x\to 2}\frac{2^{x-2}-1}{x-2}=$

(A) 1 (B) $\log_e 2$

(C) $-\log_e 2$ (D) – 1

Limit of Logarithmic Functions

Q35. $\lim_{x\to 0}\frac{\sin\left(x^{1/3}\right)\ln(1+3x)}{\left(\tan^{-1}\sqrt{x}\right)^2\left(e^{5x^{1/3}}-1\right)}$

(A) $\frac{3}{5}$ (B) $\frac{1}{5}$

(C) $\frac{2}{5}$ (D) $\frac{5}{3}$

(1)^Infinity Form

Q36. $\lim_{x\to 0}\left(\frac{\ell n(1+x)}{x}\right)^{1/x}$ equals

(A) e (B) e^{-1}
(C) e^{2} (D) $e^{-1/2}$

Q37. $\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^{x}=$

(A) e^{4} (B) e^{2}
(C) e^{3} (D) e

Q38. $\lim_{x\to\infty}\left(\frac{x}{1+x}\right)^{x}=?$

(A) e (B) e^{2}
(C) e^{-1} (D) $e^{-1/2}$

Q39. $\lim_{x\to\infty}\left(\frac{3x-4}{3x+2}\right)^{\frac{x+1}{3}}=$

(A) $e^{2/3}$ (B) e^{2}
(C) e^{-2} (D) $e^{-2/3}$

L'Hospital Rule

Q40. $\lim_{x\to 0} x\log x$ equals-

(A) 0 (B) 1
(C) 2 (D) 3

Q41. The value of $\lim_{x\to 0}\frac{x^2-2x}{2\sin x}$ is-

(A) 1 (B) 2
(C) –1 (D) 3

Q42. $\lim_{x\to 1}\frac{(2x+3)(\sqrt{x}-1)}{2x^2+x-3}$ equals-

(A) 1/4 (B) 1/2
(C) 3/4 (D) Does not exist

Q43. $\lim_{x\to 2}\frac{(\cos\theta)^x+(\sin\theta)^x-1}{x-2}$ equals-

(A) $\cos^2\theta\ \log\cot\theta$
(B) $\sin^2\theta\log\sin\theta$
(C) $\cos^2\theta$
(D) None of these

Q44. If f(a) = 2, f′ (a)= 1, g(a) = – 1, g′(a) = 2, then the value of $\lim_{x\to a}\frac{g(x)f(a)-g(a)f(x)}{x-a}$ is-

(A) –5
(B) $\frac{1}{5}$
(C) 5
(D) None of these

Q45. $\lim_{x\to 1}\frac{\sqrt[3]{x^2}-2\sqrt[3]{x}+1}{(x-1)^2}$ is equal to-

(A) $\frac{1}{9}$
(B) $\frac{1}{6}$
(C) $\frac{1}{3}$
(D) None of these

Q46. If $\lim_{x\to 0}\frac{\log(3+x)-\log(3-x)}{x}=k$, the value of k is-

(A) $-\frac{2}{3}$ (B) 0
(C) $-\frac{1}{3}$ (D) $\frac{2}{3}$

Q47. $\lim_{x\to 1}\frac{x^2-\sqrt{x}}{\sqrt{x}-1}=$

(A) 1 (B) 2
(C) 3 (D) None of these

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Let $f(x)=\begin{cases}1+\frac{2x}{a}, & 0\le x<1\\ ax, & 1\le x<2\end{cases}$. If $\lim\limits_{x\to 1} f(x)$ exists, then a is -

(A) 1 (B) – 1

(C) 2 (D) – 2

Q2. If $\lim\limits_{x\to 1}(2-x+a[x-1]+b[1+x])$ exists, then a and b can take the values.

(where [·] denotes the greatest integer function)

(A) $a=\frac{1}{3}, b=1$ (B) a = 1, b = 1

(C) a = 9, b = – 9 (D) $a=2, b=\frac{2}{3}$

Q3. If $L=\lim\limits_{x\to a}\frac{|2\sin x-1|}{2\sin x-1}$ then -

(A) limit does not exist when $a=\frac{\pi}{6}$

(B) L = – 1 when a = π

(C) L = 1 when $a=\frac{\pi}{2}$

(D) L = 1 when a = 0

Q4. Given a real-valued function f such that

$$f(x)=\begin{cases}\frac{\tan^2\{x\}}{\left(x^2-[x]^2\right)} & \text{for } x>0\\ 1 & \text{for } x=0,\\ \sqrt{\{x\}\cot\{x\}} & \text{for } x<0\end{cases}$$

where [x] is the integral part and {x} is the fractional part of x, then -

(A) $\lim\limits_{x\to 0^+} f(x)=1$

(B) $\lim\limits_{x\to 0^-} f(x)=\cot 1$

(C) $\cot^{-1}\left(\lim\limits_{x\to 0^-} f(x)\right)^2=1$

(D) $\tan^{-1}\left(\lim\limits_{x\to 0^+} f(x)\right)=\frac{\pi}{4}$

Q5. The value of $\lim\limits_{x\to\infty}\left(\frac{(2x-3)(3x+5)(4x-6)}{3x^3+x-1}\right)$ is greater than or equal to -

(A) 2 (B) 1

(C) 8 (D) 10

Q6. $\lim\limits_{x\to 1} x^{1/1-x}$ equals-

(A) 1 (B) e

(C) 1/e (D) e^2

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 7 to 9)

Let $f(x)=\frac{\sin^{-1}(1-\{x\})\times\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\times(1-\{x\})}$

where {x} denotes the fractional part of x.

$L=\lim\limits_{x\to 0^+} f(x)$

$R=\lim\limits_{x\to 0^-} f(x)$

Q7. L is equal to

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{2\sqrt{2}}$

(C) $\frac{\pi}{\sqrt{2}}$ (D) $\sqrt{2}\pi$

Q8. R is equal to

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{2\sqrt{2}}$

(C) $\frac{\pi}{\sqrt{2}}$ (D) $\sqrt{2}\pi$

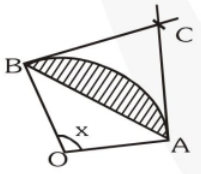
Q9. Which of the following is NOT TRUE ?

(A) $\cos L < \cos R$ (B) $\tan(2L) > \tan R$

(C) $\sin L > \sin R$ (D) None of these

Paragraph # 2 (Ques. 10 to 12)

A circular arc of radius 1 subtends an angle of x radians as shown in figure. The centre of the circle is O and the point C is the intersection of two tangent lines at A and B. Let T(x) be the area of triangle ABC and let s(x) be the area of shaded region.



Q10. $\lim_{x\to 0}\frac{T(x)}{x^3}$

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{8}$

Q11. $\lim_{x\to 0}\frac{s(x)}{x}$

(A) 0 (B) $\frac{1}{2}$

(C) 1 (D) none of these

Q12. $\lim_{x\to 0}\frac{T(x)}{s(x)}$

(A) $\frac{1}{4}$ (B) $\frac{3}{4}$

(C) $\frac{3}{2}$ (D) 0

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13.

Column-I	**Column-II**
(A) If $L = \lim_{x\to -1}\frac{\sqrt[3]{(7-x)}-2}{(x+1)}$, then $12L =$	(P) -2
(B) If $L = \lim_{x\to \pi/4}\frac{\tan^3 x - \tan x}{\cos\left(x+\frac{\pi}{4}\right)}$, then $\frac{-L}{8} =$	(Q) 2
(C) If $L = \lim_{x\to 1}\frac{(2x-3)(\sqrt{x}-1)}{2x^2+x-3}$, then $20L =$	(R) 1
(D) If $L = \lim_{x\to \infty}\frac{\log x^n - [x]}{[x]}$, where $n \in N$, then $-2L =$	(S) -1

(Note : [x] denotes greatest integer less than or equal to x)

Q14. **Column-I** **Column-II**

(A) $\lim\limits_{x\to 0}\dfrac{\sqrt{1+\tan x}-\sqrt{1+\sin x}}{x^3}=$ (P) 0

(B) $\lim\limits_{x\to 0}\dfrac{\tan(x)+2\sin^2(x)+5x^4}{3\sin(x)-x^2+x^3}=$ (Q) $\dfrac{1}{2}$

(C) $\lim\limits_{x\to\infty}\dfrac{e^{\frac{1}{x^2}}-1}{\pi-2\arctan x^2}=$ (R) $\dfrac{1}{3}$

(D) $\lim\limits_{x\to 1}\cos\left(\dfrac{\pi}{x+1}\right)\cdot\cos\left(\dfrac{\pi}{x-1}\right)=$ (S) $\dfrac{1}{4}$

Q15. **Column-I** **Column-II**

(A) $\lim\limits_{x\to\infty}\left(\dfrac{x}{1+x}\right)^x$ equals (P) e^2

(B) $\lim\limits_{x\to\infty}\left(\sin\dfrac{1}{x}+\cos\dfrac{1}{x}\right)^x$ equals (Q) $e^{-1/2}$

(C) $\lim\limits_{x\to 0}(\cos x)^{\cot^2 x}$ equals (R) e

(D) $\lim\limits_{x\to 0}\left(\tan\left(\dfrac{\pi}{4}+x\right)\right)^{1/x}$ equals (S) e^{-1}

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q16. Evaluate the following limit :

$$\lim_{x\to -7}\frac{[x]^2+15[x]+56}{\sin(x+7)\sin(x+8)}$$

where [ ] denotes the greatest integer function.

Q17. Let $f(x)=ax^3+bx^2+cx+d$ and $g(x)=x^2+x-2$.

If $\lim\limits_{x\to 1}\dfrac{f(x)}{g(x)}=1$ and $\lim\limits_{x\to -2}\dfrac{f(x)}{g(x)}=4$

then find the value of $\dfrac{c^2+d^2}{a^2+b^2}$.

Q18. If $\lim\limits_{x\to\infty}\left(\dfrac{(x+1)^{10}+(x+2)^{10}+....(x+100)^{10}}{x^{10}+10^{10}}\right)$

comes out to be 'p' then evaluate $\dfrac{p}{10}$.

Q19. If

$$\lim_{x\to\infty}\frac{a\left(2x^3-x^2\right)+b\left(x^3+5x^2-1\right)-c\left(3x^3+x^2\right)}{a\left(5x^4-x\right)-bx^4+c\left(4x^4+1\right)+2x^2+5x}=1,$$

then the value of $(a+b+c)$ can be expressed in the lowest form as $\frac{p}{q}$.

Find the value of $(p+q)$.

Q20. Let $L=\prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$; $M=\prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$

and $N=\prod_{n=1}^{\infty}\frac{\left(1+n^{-1}\right)^2}{1+2n^{-1}}$

then find the value of $L^{-1}+M^{-1}+N^{-1}$.

Q21. If $\lim_{x\to3}\left(\frac{\sqrt{2x+3}-x}{\sqrt{x+1}-x+1}\right)^{\frac{x-1-\sqrt{x^2-5}}{x^2-5x+6}}$ can be expressed in the form $\frac{a\sqrt{b}}{c}$ where $a, b, c \in N$, then find the least value of $(a^2+b^2+c^2)$.

SUBJECTIVE TYPE

Q22. $\lim_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}$

Q23. $\lim_{x\to\infty}\frac{(2x+1)^{40}(4x-1)^5}{(2x+3)^{45}}$

Q24. Evaluate $\lim_{x\to\infty}\sqrt{x^2+x+1}-\sqrt{x^2+1}$

Q25. $\lim_{n\to\infty}\left(3^{1/2}\cdot3^{1/4}\cdot3^{1/8}\ldots.3^{1/2^n}\right)$ equals-

Q26. If $\lim_{x\to\infty}\left(\frac{x^2+1}{x+1}-ax-b\right)=0$ then find values of a & b.

Q27. $\lim_{n\to\infty}\frac{(n+2)!+(n+3)!}{(n+4)!}$

Q28. $\lim_{x\to\tan^{-1}3}\frac{\tan^2x-2\tan x-3}{\tan^2x-4\tan x+3}$

Q29. Evaluate $\lim_{x\to0}\frac{e^{x^2}-\cos x}{x^2}$

Q30. Evaluate $\lim_{x\to\pi/2}\frac{a^{\cot x}-a^{\cos x}}{\cot x-\cos x}$

Q31. Evaluate $\lim_{x\to-1}\frac{\sqrt{x+2}-1}{\log_5(x+2)}$

Q32. Evaluate $\lim_{x\to0}\frac{\left(2^x-1\right)^3}{\sin\frac{x}{4}\log\left(1+\frac{x^2}{2}\right)}$

Q33. Evaluate the following limits

(i) $\lim_{x\to\infty}\left(\frac{x+5}{x-1}\right)^x$

(ii) $\lim_{x\to\infty}\left(\frac{x^2+4x-3}{x^2-2x+5}\right)^x$

Q34. Evaluate : $\lim_{x\to a}\frac{x^a-a^x}{x^x-a^a}$

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. $\lim\limits_{x\to5}\dfrac{x^2-9x+20}{x-[x]}=$; where [.] denotes greatest integer function :-

(A) 0 (B) 1

(C) Does not exist (D) None

Q2. $\lim\limits_{x\to0^-}\sin^{-1}[\tan x]=\ell$ then $\{\ell\}$ is equal to Where [ ] and { } denotes integer and fractional part :-

(A) 0 (B) $1-\dfrac{\pi}{2}$

(C) $\dfrac{\pi}{2}-1$ (D) $2-\dfrac{\pi}{2}$

Q3. $\lim\limits_{x\to0}\dfrac{\sin[\cos x]}{1+[\cos x]}$

([.] denotes the greatest integer function)

(A) equal to 1 (B) equal to 0

(C) does not exist (D) None of these

Q4. $\lim\limits_{n\to\infty}\dfrac{1-2+3-4+5-6+\ldots-2n}{\sqrt{n^2+1}+\sqrt{4n^2-1}}$ is equal to

(A) $\dfrac{1}{3}$ (B) $-\dfrac{1}{3}$

(C) $-\dfrac{1}{5}$ (D) Does not exist

Q5. Let $f(x)=\lim\limits_{n\to\infty}\dfrac{x^{2n}-1}{x^{2n}+1}$ then :-

(A) f(x) = 1; for |x| > 1

(B) f(x) = 1; for |x| < 1

(C) f(x) = 1; for |x| = 1

(D) f(x) is not defined for any value of x

Q6. $\lim\limits_{x\to0}\dfrac{1-\cos^3 x}{x\sin x\cos x}$ equals-

(A) 1/4 (B) 1

(C) 3/2 (D) Does not exist

Q7. $\lim\limits_{x\to0}\dfrac{\tan x-\sin x}{x^3}$ is-

(A) 0 (B) $\dfrac{1}{2}$

(C) 2 (D) none of these

Q8. $\lim\limits_{x\to0}\dfrac{\sin\left(\pi\cos^2 x\right)}{x^2}$ equals-

(A) $-\pi$ (B) π

(C) $\pi/2$ (D) 1

Q9. $\lim\limits_{x\to\infty}(\sin\sqrt{x+1}-\sin\sqrt{x})$ is equal to -

(A) 1 (B) –1

(C) 0 (D) none of these

Q10. Let $r=\lim\limits_{x\to0}\left(\dfrac{\sin x}{x}\right)^{\frac{1}{x}}$ and $m=\lim\limits_{x\to0}\left(\dfrac{\sin x}{x}\right)^{\frac{1}{x^2}}$ then -

(A) $r=0$ and $m=e^{1/6}$

(B) $r=1$ and $m=e^{1/6}$

(C) $r=0$ and $m=e^{-1/6}$

(D) $r=1$ and $m=e^{-1/6}$

Q11. The value of $\lim\limits_{x\to\infty}x\left(\tan^{-1}\dfrac{x+1}{x+2}-\dfrac{\pi}{4}\right)$ is

(A) $\dfrac{1}{2}$ (B) $-\dfrac{1}{2}$

(C) 1 (D) –1

Q12. $\lim\limits_{x\to\pi/2}\dfrac{\cot x-\cos x}{(\pi-2x)^3}$

(A) $\dfrac{1}{16}$ (B) $-\dfrac{1}{16}$

(C) $\dfrac{1}{8}$ (D) $-\dfrac{1}{8}$

Q13. $\lim\limits_{x\to 0}\dfrac{\left[\dfrac{x}{\sin x}\right]-\left[\dfrac{\sin x}{x}\right]}{\left[\dfrac{\tan x}{x}\right]}$ is equal to ([.] represents the greatest integer function) -

(A) 1 (B) –1

(C) 0 (D) does not exists

Q14. $\lim\limits_{x\to \pi/2}\dfrac{\left[1-\tan\left(\dfrac{x}{2}\right)\right][1-\sin x]}{\left[1+\tan\left(\dfrac{x}{2}\right)\right][\pi-2x]^3}$ is-

(A) ∞ (B) $\dfrac{1}{8}$ (C) 0 (D) $\dfrac{1}{32}$

Q15. The value of $\lim\limits_{x\to 0}\dfrac{\sin^{-1}(2x)-\tan^{-1}(x)}{\sin x}$ is equal to-

(A) 0 (B) $\dfrac{1}{2}$

(C) 1 (D) –1

Q16. Let $f(x)=\lim\limits_{n\to\infty}\dfrac{\sin\left(\pi x^4\right)+(x+2)^n\cdot\dfrac{\tan(\pi x)}{x+1}}{1+(x+2)^n-x^4}$,

then $\lim\limits_{x\to -1} f(x)$ is equal to -

(A) π (B) $\dfrac{22}{7}$

(C) 1 (D) $\pi-\dfrac{22}{7}$

Q17. $\lim\limits_{x\to\infty}(\sqrt{x+\sqrt{x+\sqrt{x}}}-\sqrt{x})$ equals-

(A) 0 (B) 1/2

(C) log2 (D) e^4

Q18. If $\lim\limits_{x\to 0}\dfrac{x(1+a\cos x)-b\sin x}{x^3}=1$ then the value of a and b are respectively-

(A) 1/2, 3/2 (B) 3/2, 5/2

(C) –5/2, –3/2 (D) None of these

Q19. $\lim\limits_{x\to 2}\dfrac{\sin\left(e^{x-2}-1\right)}{\log(x-1)}$ equals-

(A) 0 (B) 1 (C) –1 (D) 2

Q20. $\lim\limits_{n\to\infty}\left(\dfrac{1}{3.5.7.9}+\dfrac{1}{5.7.9.11}+\ldots\ldots\ldots+\ldots\ldots\right.$

$\left.+\dfrac{1}{(2n+1)(2n+3)(2n+5)(2n+7)}\right)=$

(A) $\dfrac{1}{630}$ (B) $\dfrac{1}{105}$

(C) $\dfrac{1}{315}$ (D) $\dfrac{1}{660}$

Q21. The value of $\lim\limits_{x\to 0}\dfrac{27^x-9^x-3^x+1}{\sqrt{5}-\sqrt{4+\cos x}}$ is :-

(A) $\sqrt{5}(\log 3)^2$ (B) $8\sqrt{5}\log 3$

(C) $16\sqrt{5}\log 3$ (D) $8\sqrt{5}(\log 3)^2$

Q22. $\lim\limits_{x\to 0}\left(\dfrac{1^x+2^x+3^x+4^x}{4}\right)^{\frac{4}{x}}=$

(A) $(24)^{\frac{1}{4}}$ (B) 24

(C) $\dfrac{1}{24}$ (D) $\left(\dfrac{1}{24}\right)^{\frac{1}{4}}$

Q23. If $f(x)=\lim\limits_{t\to 0}\tan^{-1}\left(\dfrac{e^{xt}-1}{t}\right)$

then $\lim\limits_{x\to 0}\dfrac{f(x)-x}{x^3}$ is equal to

(A) $\dfrac{1}{2}$ (B) $\dfrac{-1}{2}$

(C) $\dfrac{1}{3}$ (D) $\dfrac{-1}{3}$

Q24. If a, b, c, d are positive, then

$\lim_{x\to\infty}\left(1+\frac{1}{a+bx}\right)^{c+dx}$ is equal to

(A) $e^{d/b}$ (B) $e^{c/a}$

(C) $e^{(c+d)/a+b}$ (D) e

Q25. If $\lim_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)^{2x}=e^2$, then the values of a and b, are-

(A) $a \in R, b \in R$

(B) $a = 1, b \in R$

(C) $a \in R, b = 2$

(D) a = 1 and b = 2

Q26. If $\lim_{x\to0}\frac{\sin 2x+a\sin x}{x^3}$ be finite, then the value of a and the limit are given by-

(A) –2, 1 (B) –2, –1

(C) 2, 1 (D) 2, –1

Q27. Let $f(x)=\frac{1}{\sqrt{18-x^2}}$, then the value of

$\lim_{x\to3}\frac{f(x)-f(3)}{x-3}$ is

(A) 0 (B) –1/9

(C) –1/3 (D) 1/9

Q28. Let f(x) be a twice differentiable function and f"(0) = 5.

Then $\lim_{x\to0}\frac{3f(x)-4f(3x)+f(9x)}{x^2}$ is equal to-

(A) 30 (B) 120

(C) 40 (D) none of these

Q29. $\lim_{x\to\pi/2}\tan x\ \ell n\sin x$ equals

(A) 0 (B) 1

(C) – 1 (D) None of these

Q30. $\lim_{x\to0}\frac{\sqrt[3]{1+3x}-1-x}{(1+x)^{101}-1-101x}$ has the value equal to -

(A) $-\frac{3}{5050}$ (B) $-\frac{1}{5050}$

(C) $\frac{1}{5051}$ (D) $\frac{1}{4950}$

Q31. If f(1) = 1, f'(1) = 2, then $\lim_{x\to1}\frac{\sqrt{f(x)}-1}{\sqrt{x}-1}=$

(A) 2 (B) 1 (C) 3 (D) 4

Q32. Let f(a) = g(a) = k and their n^{th} derivatives $f^n(a)$, $g^n(a)$ exist and are not equal for some n. Further if

$\lim_{x\to a}\frac{f(a)g(x)-f(a)-g(a)f(x)+g(a)}{g(x)-f(x)}=4$

then the value of k is-

(A) 0 (B) 4 (C) 2 (D) 1

Q33. $\lim_{x\to0}\frac{\sin(6x^2)}{\ell n\cos(2x^2-x)}=$

(A) 12 (B) –12 (C) 6 (D) –6

Q34. $\lim_{n\to\infty}\frac{\{x\}+\{2x\}+\{3x\}+...+\{nx\}}{n^2}=$ where { . } denotes fractional function

(A) 1 (B) 0

(C) 1/2 (D) None

Q35. $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r}{n^2+n+r}=?$

(A) 0 (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) 1

Q36. $\lim_{n\to1}\sum_{x=1}^{10}\frac{1-\cos^{2n}(x-11)}{(x-11)^2}$ is equal to :-

(A) 0 (B) 5

(C) 10 (D) None of these

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $L=\lim_{x\to a}\frac{\sqrt{3x^2+a^2}-\sqrt{x^2+3a^2}}{x-a}$, then -

(A) L = 1, for all $a\in R$

(B) L = 1, for all $a>0$

(C) L = – 1, for $a<0$

(D) L does not exist for a = 0

Q2. Let $f:\left(\frac{-\pi}{2},\frac{\pi}{2}\right)\to R$, be defined and as f(x)

$$=\begin{cases}\lim\limits_{n\to\infty}\left(\frac{(\tan x)^{2n}+x^2}{\sin^2 x+(\tan x)^{2n}}\right), & x\neq 0\\ 1, & x=0\end{cases}$$. Which of the following holds good?

(A) $f\left(\frac{\pi^-}{4}\right)=f\left(\frac{\pi^+}{4}\right)$

(B) $f\left(\frac{-\pi^-}{4}\right)=f\left(\frac{\pi^+}{4}\right)$

(C) $f\left(\frac{-\pi^+}{4}\right)=f\left(\frac{\pi^-}{4}\right)$

(D) $f(0^+)=f(0^-)=f(0)$

Q3. Let $L=\lim_{x\to 0}\frac{3px+(p-2)\sin x}{(\sin^{-1}x)^3}$, where $p\in R$.

If L is finite then -

(A) $p=\frac{1}{2}$ (B) $L=\frac{1}{4}$

(C) $L=\frac{-1}{4}$ (D) $p=\frac{-1}{2}$

Q4. Let $f(x)=(e^x+2x)^{\frac{3}{x}}$, then

(A) $\lim\limits_{x\to 0}f(x)=9$ (B) $\lim\limits_{x\to\infty}f(x)=e^3$

(C) $\lim\limits_{x\to 0}f(x)=e^9$ (D) $\lim\limits_{x\to\infty}f(x)=3$

Q5. If $\lim_{x\to 0}\frac{ax^2+bx+c}{\sin x}=3$ then (a + b + c) may be

(A) 0 (B) 3

(C) 30 (D) 33

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 6 to 8)

Let m, n are non zero integers and

$\lim_{x\to 0}\frac{\tan mx-n\sin x}{x^3}$ = an integer.

Q6. Which of the following statement is true –

(A) m is should be an even but n is odd

(B) both m & n should be odd

(C) m is odd and n is even

(D) both m & n are even integers

Q7. The value of limit in terms of m & n is –

(A) $\frac{2m+n^2}{6}$ (B) $\frac{2m^3+n}{6}$

(C) ∞ (D) None of these

Q8. Is m & n are related as –

(A) $m^2=n$ (B) $m=n^2$

(C) m = n (D) None of these

Paragraph # 2 (Ques. 9 to 11)

If f, g and h are functions having a common domain D and $h(x) \le f(x) \le g(x)$, $x \in D$ and if $\lim_{x\to a} h(x) = \lim_{x\to a} g(x) = l$ then $\lim_{x\to a} f(x) = l$. This is known as Sandwich Theorem. Using this result, compute the following limits.

Q9. The value of $\lim_{x\to 0} \frac{|x|}{\sqrt{x^4+4x^2+7}}$

(A) 1 (B) 0

(C) 1/2 (D) does not exist

Q10. $x^4 \sin\left(\frac{1}{3\sqrt{x}}\right)$ is -

(A) 0 (B) 1

(C) 1/3 (D) does not exist

Q11. Let $f(x) = x^2 \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}}$, $x \neq 0$ and $f(0) = 1$ then -

(A) $\lim_{x\to 0^+} f(x)$ doesn't exist

(B) $\lim_{x\to 0} f(x)$ doesn't exist

(C) f(x) exist

(D) $\lim_{x\to 0} f(x) = 1$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q12.

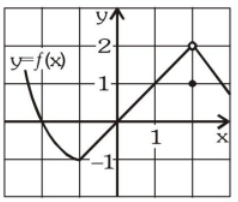
Column-I	**Column-II**
(A) $\lim_{x\to 0^+}\left(\frac{x+1}{2x^2+3x+5}\right)^{1/x}$	(P) e
(B) $\lim_{x\to 0}\left(\left\{\frac{\sin x}{x}\right\}\right)^{-\frac{6}{x^2}}$	(Q) 0
(C) $\lim_{x\to 0}\left(\left[\frac{\tan x}{x}\right]\right)^{\frac{1}{x^2}}$	(R) 1
(D) $\lim_{x\to 0^+}\left(2^{1/x}+3^{1/x}+5^{1/x}\right)^x$	(S) 5

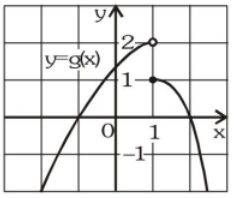
Note : where [k] and {f} denotes greatest integer function and fractional part function.

Q13.

Column-I	**Column-II**
(Note : [·] denotes the greatest integer function)	
(A) $\lim_{x\to 0}\left(\left[100\frac{\sin x}{x}\right]+\left[100\frac{\tan x}{x}\right]\right) =$	(P) 198
(B) $\lim_{x\to 0}\left(\left[100\frac{x}{\sin x}\right]+\left[100\frac{\tan x}{x}\right]\right) =$	(Q) 199
(C) $\lim_{x\to 0}\left(\left[100\frac{\sin^{-1} x}{x}\right]+\left[100\frac{\tan^{-1} x}{x}\right]\right) =$	(R) 200
(D) $\lim_{x\to 0}\left(\left[100\frac{x}{\sin^{-1} x}\right]+\left[100\frac{\tan^{-1} x}{x}\right]\right) =$	(S) 201

Q14. The graphs of f and g are given. Use them to evaluate each limit





Column-I	**Column-II**
(A) $\lim_{x\to 1} f(g(x))$	(P) 1
(B) $\lim_{x\to 2} \sqrt{3f(x)-2}$	(Q) does not exist
(C) $\lim_{x\to 0} \frac{f(x)}{g(x)} + f(x)g(x)$	(R) 0
(D) $\lim_{x\to 1^+} \frac{3f(x)-g(x)}{f(x)+g(x)}$	(S) 2

Q15.

Column-I	**Column-II**
(A) $\lim_{n\to\infty} n^{-n^2}\left\{(n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{2^2}\right)\ldots\left(n+\frac{1}{2^{n-1}}\right)\right\}^n$	(P) $e^{-1/6}$
(B) $\lim_{x\to\infty} \left(\frac{x^2+4x-3}{x^2-2x+5}\right)^x$	(Q) e
(C) $\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{1/x^2}$	(R) e^6
(D) $\lim_{x\to\infty} \left(\sin\frac{1}{x}+\cos\frac{1}{x}\right)^x$	(S) e^2

TRIPLE MATCH COLUMN TYPE

This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. ***ONLY ONE*** *of these four is correct.*

Column I	**Column II**	**Column III**
(I) $\lim_{x\to 0} \frac{\sin ax+bx}{x^3} = 36$	(i) a = – 5/2	(P) b = –1
(II) $\lim_{x\to 0} \frac{x(1+a\cos x)-b\sin x}{x^3} = 1$	(ii) a = – 6	(Q) b = – 3/2
(III) $\lim_{x\to 0} \frac{(4x-1)^{1/3}+a+bx}{x} = \frac{1}{3}$	(iii) a = 2	(R) b = 6
(IV) $\lim_{x\to 0} \frac{\sqrt{ax+b}-2}{x} = \frac{1}{2}$	(iv) a = 1	(S) b = 4

Q16. Which of the following options is the only CORRECT combination?

(A) (I) (ii) (R) (B) (II) (iii) (P)

(C) (III) (ii) (S) (D) (IV) (i) (Q)

Q17. Which of the following options is the only CORRECT combination?

(A) (IV) (i) (P) (B) (II) (i) (Q)

(C) (I) (i) (Q) (D) (III) (iv) (S)

Q18. Which of the following options is the only CORRECT combination?

(A) (I) (iv) (P) (B) (II) (iv) (Q)

(C) (III) (iv) (P) (D) (IV) (iv) (S)

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. If a, b, c and d are real constants such that $\lim\limits_{x\to 0}\dfrac{ax^2+\sin bx+\sin cx+\sin dx}{3x^2+5x^4+7x^6}=8$, then find the value of (a + b + c + d).

Q20. Let $\lim\limits_{x\to 1}\dfrac{x^6+1+\frac{5}{3}(x^4+x^2)-\frac{8}{3}(x^5+x)}{x^5+x+6x^3-4x^4-4x^2}=\dfrac{p}{q}$ where p, q ∈ N.

Find the smallest value of (p + q).

Q21. If $\lim\limits_{n\to\infty}\dfrac{n.3^n}{n(x-2)^n+n.3^{n+1}-3^n}=\dfrac{1}{3}$ where n∈N, then the number of integer(s) in the range 'x' is -

Q22. If $f(x)=\lim\limits_{n\to\infty}\dfrac{2}{n^2}\left(\sum\limits_{k=1}^{n}kx\right)\cdot\left(\dfrac{3^{nx}-1}{3^{nx}+1}\right)$ where n ∈ N, then find the sum of all the solutions of the equation $f(x)=|x^2-2|$.

Q23. Suppose ABC is a triangle such that ∠BAC is a right angle. A point H is located at BC such that AH is perpendicular to BC. If ∠ABC = θ, AB = a, and the value of $\lim\limits_{\theta\to 0}\dfrac{AC+AH}{\theta}$ = 10, then the value of a is equal to -

Q24. At the end-points and the midpoint of a circular arc AB tangent lines are drawn, and the points A and B are joined with a chord. Prove that the ratio of the areas of the two triangles thus formed tends to 4 as the arc AB decreases indefinitely.

Q25. Let $\lim\limits_{x\to\infty}x\ln\left(e\left(1+\dfrac{1}{x}\right)^{1-x}\right)$ equal $\dfrac{m}{n}$ where m and n are relatively prime positive integer. Find (m + n).

SUBJECTIVE TYPE

Q26. Without using L hospital, expansion

Evaluate : $\lim\limits_{x\to 0}\dfrac{x-\sin x}{x^3}$

Q27. $\lim\limits_{x\to -1}\dfrac{\cos 2-\cos 2x}{x^2-|x|}$ equals-

Q28. Evaluate : $\lim\limits_{x\to 0}\dfrac{\tan[-2\pi^2]x^2-x^2\tan[-2\pi^2]}{\sin^2 x}$

Q29. The value of $\lim\limits_{x\to a}\dfrac{\log(x-a)}{\log(e^x-e^a)}$ is :-

Q30. Evaluate : $\lim\limits_{x\to 0}\left(\dfrac{a^x+b^x+c^x}{3}\right)^{1/x}$

Q31. Through a point A on a circle, a chord AP is drawn & on the tangent at A a point T is taken such that AT = AP. If TP produced meet the diameter through A at Q, prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.

Q32. At the end points A, B of the fixed segment of length L, lines are drawn meeting in C and making angles θ and 2θ respectively with the given segment. Let D be the foot of the altitude CD and let x represents the length of AD. Find the value of x as θ tends to zero i.e. $\lim_{\theta\to 0} x$.

Q33. Evaluate $\lim_{x\to\infty}\frac{\log x^n-[x]}{[x]}$, where $n \in N$ and [.] is greatest integer function.

Q34. Let $f(x)=\lim_{n\to\infty}\frac{2x^{2n}\sin\frac{1}{x}+x}{1+x^{2n}}$ then find -

(i) $\lim_{x\to\infty} x\,f(x)$

(ii) $\lim_{x\to 1} f(x)$

(iii) $\lim_{x\to 0} f(x)$

(iv) $\lim_{x\to-\infty} f(x)$

Answer Key

Ex - 1

1	2	3	4	5	6	7	8	9	10
A	D	D	C	C	C	A	A	D	B
11	12	13	14	15	16	17	18	19	20
A	D	D	B	D	A	B	A	B	A
21	22	23	24	25	26	27	28	29	30
C	C	B	C	B	D	A	B	A	C
31	32	33	34	35	36	37	38	39	40
D	D	C	B	A	D	A	C	D	A
41	42	43	44	45	46	47			
C	B	D	C	A	D	C			

Ex - 1A

1. BC **2.** BC **3.** ABC **4.** ACD **5.** C

6. C **7.** A **8.** B **9.** D **10.** D

11. A **12.** C **13.** (A) → S; (B) → R; (C) → P; (D) → Q

14. (A) → S; (B) → R; (C) → Q; (D) → P **15.** (A) S; (B) R; (C) Q; (D) P

16. 0, **17.** 16 **18.** 10 **19.** 167 **20.** 8

21. 29 **22.** $\frac{2}{3\sqrt{3}}$ **23.** 32 **24.** $\frac{1}{2}$ **25.** 3

26. a=1 & b= –1 **27.** 0 **28.** 2 **29.** $\frac{3}{2}$ **30.** $\log_e a$

31. $\frac{1}{2}\log_e 5$ **32.** $8(\log_e 2)^3$ **33.** (i) e^6 (ii) Sol not available **34.** $\frac{1-\log a}{1+\log a}$

Ex - 2									
1	2	3	4	5	6	7	8	9	10
C	D	B	B	A	C	B	B	C	D
11	12	13	14	15	16	17	18	19	20
B	A	A	D	C	A	B	C	B	A
21	22	23	24	25	26	27	28	29	30
D	B	D	A	B	B	D	B	A	B
31	32	33	34	35	36				
A	B	B	B	C	D				

Ex - 2A

1. B,C,D **2.** B,C,D **3.** A,B **4.** B,C **5.** A,B,C,D

6. D **7.** B **8.** C **9.** B **10.** A

11. C

12. (A) $\rightarrow$ Q; (B) $\rightarrow$ P; (C) $\rightarrow$ R; (D) $\rightarrow$ S

13. (A) $\rightarrow$ Q; (B) $\rightarrow$R; (C) $\rightarrow$ Q; (D) $\rightarrow$ P

14. (A) $\rightarrow$ Q; (B) $\rightarrow$ S; (C) $\rightarrow$ R; (D) $\rightarrow$ P

15. (A) $\rightarrow$ (S); (B) $\rightarrow$ (R); (C) $\rightarrow$ (P); (D) $\rightarrow$ (Q)

16. A **17.** B **18.** C **19.** 24 **20.** 0013

21. 5 **22.** 0 **23.** 5 **24.** 4 **25.** 0005

26. $\frac{1}{6}$ **27.** $2\sin 2$ **28.** $-20 + \tan 20$ **29.** 1 **30.** $(abc)^{1/3}$

31. 4r **32.** $\frac{2L}{3}$ **33.** –1 **34.** (i) 2, (ii) DNE, (iii) 0, (iv) 0

-: Important Notes & Formulas :-